Repo link https://github.com/HarrisonKaufman/virtrade

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2. Paper King$
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4. Application Name - Virtrade

5. Our application is going to be a paper trading software focused around improving your stock trading abilities to increase real profits. This will feature several things like the ability to purchase and sell stocks real time based on the NYSE and also track valuable metrics related to your trading. Since this will be linked to an external API, users will be practicing in an environment that closely mimics real stock trading.

   We aim to provide a sandbox environment to allow beginner traders to test various strategies and analysis techniques in a more immersive, realistic, and approachable manner than traditional backtesting.

6. Our target audience will be newcomers to the stock market who are looking to get more skilled in trading, while using fake money. We aim to help new users and newcomers not lose their own money but rather gain skills in swing trading and reading charts and candles.

7. To give beginner traders a place to master their skills and compete with large players in the NYSE.
   Unlike established paper trading platforms like Webull, our product is geared towards beginners seeking a gamified experience to build intuition.

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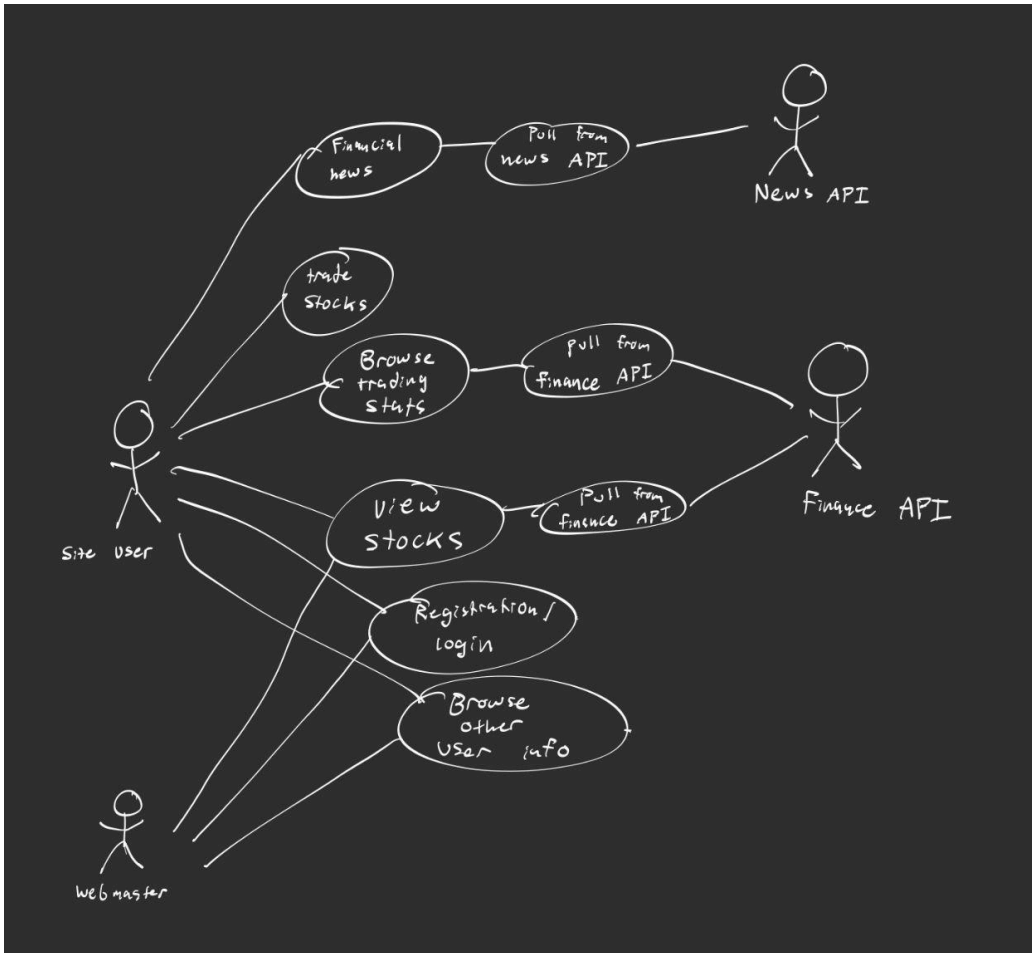
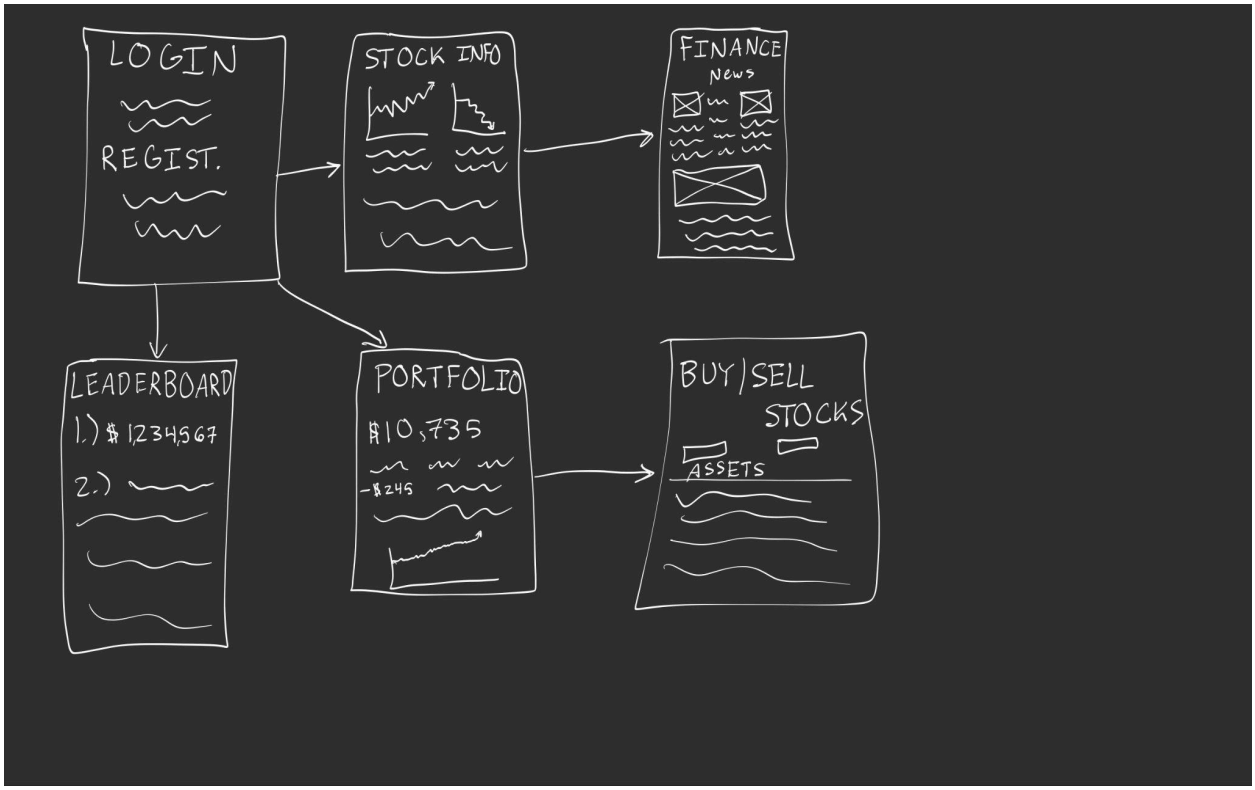
9. We will use AGILE methodology to ensure we stay on track and have a flexible plan throughout the 4 weeks. If needed, we can structure our work in sprints and assign a scrum master.

10. We will use Discord to communicate during development.
    Discord usernames:
    - dylanspektor
    - ajwatt99
    - mihai_guy
    - jumbonorman
    - silttt

11. Our planned weekly meeting time will be Tuesday 1:45-2 with our TA, the modality will be remote.
    We will have weekly meetings 2:00-2:25pm Thursday in the Engineering center to check in on our progress. These are our formal meetings but we will be regularly messaging throughout the week and calling to speak as necessary so this will be more than enough.
12. 

13. 

Extra credit risk analysis:

1. Financial regulations: We need to make sure that it is clear the environment is a sandbox environment and that we are not implementing any features that violate regulations. However, we are likely safe from these things since there is no real money involved, so it has a low severity and only requires us to educate ourselves to mitigate any regulation risks.
2. Exchange APIs: We need to make sure that the exchange we use to get stock data from has an API suitable for our needs. This means that the free request rates are acceptable for our implementation and that the data we are getting is accessible and easily parseable. This is crucial to our project so is a moderate to high risk, but we can mitigate the risk of API inadequacy by researching options and limiting rates to mitigate high load.
3. Breaking Changes: A team member could accidentally push/merge changes to main without realizing unintended consequences. We have set up branch protection rules for main on GitHub which require contributors to create a pull request before merging, and mandate that each pull request gets two approvals. This will largely mitigate the issue so it is at low severity.
4. Time series storage: Our project likely requires us to store stock price data as a time series, which could be difficult to implement in our database. However, we also know databases are flexible for a variety of data, so this is a low to moderate risk. This can be mitigated by using a postgres extension for time series data, which we will definitely need to learn about before implementing.

5. Scope creep: The scope of our project slowly grows over the course of four weeks since there is a lot we could possibly implement in terms of financial metrics and analyses. However, this quickly gets out of hand and we have far too many features to implement while losing sight of the project as a whole. As long as we are aware of this risk and stay on track with our plan, this should be low severity and largely mitigated by responsible expectations and teamwork.